

A Minor Project Report

On

**CLASSIFYING FASHION IMAGES WITH MACHINE
LEARNING**

Submitted in the Partial Fulfillment of the Requirements

For the Award of the Degree of

Bachelor of Technology

In

Computer Science Engineering

(Artificial Intelligence and Machine Learning)

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DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

B V RAJU INSTITUTE OF TECHNOLOGY

**(UGC Autonomous, Accredited by NBA & NAAC) Vishnupur, Narsapur, Medak,
Telangana State, India – 502 313**

2024-2025

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DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

CERTIFICATE

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In Partial Fulfillment of the Requirements for the Award of Degree of Bachelor of Technology in CSE (Artificial Intelligence and Machine Learning) to B V Raju Institute of Technology is Record of Bonafide Work Carried Out During the Period from December 2024 to May 2025 by Them Under the Supervision of

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CANDIDATE'S DECLARATION

We Hereby Certify that the Work Which is Being Presented in the Minor Project Entitled “**Classifying Fashion Images with Machine Learning**” in Partial Fulfillment of the Requirements For the Award of Degree of Bachelor of Technology and Submitted in the Department of CSE(Artificial Intelligence and Machine Learning), B V Raju Institute of Technology, Narsapur, is an Authentic Record of Our Own Work Carried Out During the Period From December 2024 to May 2025, Under the Supervision of **Mrs. B Divya, Assistant professor** and **Dr. G Venu Gopal ,Assistant Professor**.

The Work Presented in this Mini Project Report Has Not Been Submitted by Us for the Award of Any Other Degree of This or Any Other Institute/University.

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ABSTRACT

An individual's unique interests, attributes, and attitude can be determined from their clothing choices. When making a purchase, consumers consider both the overall attractive appearance and the clothing's functionality. Hence choosing a correct clothing for themselves is an important factor in presenting themselves. This project identifies the type of clothing it is by using Convolutional Neural Networks (CNNs), a Machine Learning technique most frequently used for image analysis. The main components of a convolutional neural network are an input layer, an output layer, and a hidden layer, which may have multiple layers. The dataset that is being used is the MobileNetV2 dataset consisting of different clothing items that are divided into training and test datasets. The model is constructed using Convolutional Neural Networks (CNN) in Keras. The CNN classifies the different types of clothing and determines which class of clothing item it belongs.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	FULL FORM
CNN	Convolutional Neural Network
CRF	Conditional Random Fields
DNN	Deep Neural Network
FCL	Fully Connected Layer
GAN	Generative Adversarial Network
GCN	Graph Convolutional Network
H-CNN	Hierarchical Convolutional Neural Network
LSTM	Long Short-Term Memory
MNIST	Modified National Institute of Standard and Technology
PS-Net	Pyramid Scene Parsing Network
R-CNN	Region Based Convolutional Neural Network
ReLU	Rectified Linear Unit
SE-Net	Squeeze and Excitation Network
SVM	Support Vector Machine
TCN	Temporal Convolutional Network

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

The fast-paced digital evolution of the fashion industry has generated a pressing need for automated image categorization systems, fueled by limitations of time-consuming manual tagging processes that are not only expensive, running into about \$0.03–\$0.10 per image for retailers, but also susceptible to human errors with misclassification rates of 5–8%, and inherently unscalable for large and dynamic catalogs.

With e-commerce sites, social commerce, and AI-based styling solutions increasing, the drawbacks of conventional approaches become more and more evident, and strong machine learning solutions are needed that can lower operational expenses more than 90%, attain human-level accuracy (94–97%), and handle thousands of images in real-time.

Computer vision has been a game-changing technology in fashion retail, driving visual search and recommendation tools employed by sites such as Pinterest, ASOS, and Zalando that allow customers to find similar products based on image queries.

Aside from client-facing tools, auto-classification simplifies inventory and supply chain operations by enabling efficient warehouse sorting and the obviation of manual metadata input, while optimizing augmented reality experiences such as virtual try-ons and AI-driven styling assistance.

But traditional deep networks like ResNet and VGG, though accurate, tend to be computationally intensive, slow to perform real-time inference, and not feasible for use on mobile or edge hardware because of their huge model size.

This is where MobileNetV2 excels as an optimized implementation, using depth wise separable convolutions and inverted residual building blocks to significantly minimize computational overhead while not compromising on accuracy.

With a model size of less than 20MB, MobileNetV2 is suitable for deployment on smartphones, IoT devices, and browser applications and hence suitable for both big-box retailers and small enterprises. Additionally, with pre-training on ImageNet, transfer learning is made efficient with high fashion-specific dataset accuracy with little fine-tuning on Fashion MNIST.

The wider industry consequence is important, as embracing MobileNetV2 for fashion categorization can make e-commerce more searchable, optimize inventory automation, and support novel customer experiences such as real-time personalization.

As the fashion world keeps changing with sustainable fashion and on-demand production, efficient and scalable AI models such as MobileNetV2 will be central in defining the future of retail technology as they provide the perfect balance of speed, precision, and deployability that cannot be replicated by older methods.

1.2 INTRODUCTION TO DOMAIN

Machine learning (ML) is a revolutionary field of artificial intelligence that allows computers to learn from data without being programmed. Through the recognition of patterns in huge sets of data, ML algorithms can predict, categorize information, and facilitate decision-making with astounding precision.

From streaming services' personalized recommendations to financial institutions' identification of fraud, ML drives many of the technologies we use on a daily basis.

At its fundamental level, machine learning is based on three paradigms: “**supervised learning**”, where models are trained using labeled data (e.g., detecting spam); “**unsupervised learning**”, which discovers latent patterns in unlabeled data (e.g., segmenting customers); and “**reinforcement learning**”, where algorithms learn by trial and error (e.g., autonomous vehicles).

Developments in deep learning, especially neural networks, have further pushed ML's boundaries, facilitating achievements in image

recognition, natural language understanding, and autonomous systems.

The advent of big data and high-performance computing has propelled the adoption of ML in all industries, ranging from healthcare (predicting diagnosis) to finance (algorithmic trading) to retail (demand forecasting). Bias reduction, model explainability, and data privacy, though, remain of vital concern.

With ongoing advancements in ML, its ability to address intricate real-world challenges surges exponentially, placing it among the most thrilling and influential areas of contemporary technology.

1.3 PROBLEM STATEMENT

Manual fashion image classification is inefficient, time-consuming, and prone to errors for online retailers and e-commerce websites. Current approaches are inaccurate, limited in scalability, and cannot support real-time processing.

This project implements an automatic fashion image classifier based on MobileNetV2 to classify fashion images efficiently and accurately, lessening human effort, enhancing search capabilities, inventory management, and customer experience in digital fashion retailing.

1.4 OBJECTIVE

This project will create a precise and fast fashion image classification system based on MobileNetV2 deep learning architecture.

The major objectives are to attain >94% accuracy in classification on the Fashion MNIST dataset, achieving optimal model performance for real-time processing, and designing a scalable solution deployable on various e-commerce platforms.

The system will computerize product categorization, improve search and browsing capabilities, and curtail manual tagging activities with minimal lightweight computational requirements optimal for mobile and edge device applications in retail contexts.

CHAPTER 2

REVIEW OF LITERATURE

2.1 INTRODUCTION

The domain of fashion image classification has seen tremendous progress through multiple machine learning strategies, as presented in recent literature. Initial research mainly focused on conventional computer vision methodologies such as SIFT and SURF feature descriptors in conjunction with SVM classifiers, with poor accuracy scores ranging from 65-75% on simple datasets.

The game-changer emerged when deep learning architectures, specifically Convolutional Neural Networks (CNNs), were adopted with better performance automatically learning discriminative features from raw pixels.

Latest literature spotlights the impressive performance of transfer learning methods in fashion classification problems. Experiments across different pre-trained models (VGG16, ResNet50, and MobileNetV2) repeatedly demonstrate that these models have 90-95% accuracy on test sets such as Fashion MNIST and DeepFashion. MobileNetV2 is especially promising with its light architecture utilizing depth wise separable convolutions and inverted residuals that deliver high accuracy at much lower computational needs.

A number of researchers have studied particular problems in fashion categorization, such as coping with large intra-class variations (e.g., various types of shirts) and inter-class similarities (e.g., shirts versus t-shirts). Methods such as attention mechanisms, triplet loss functions, and metric learning have been suggested to overcome these problems. Data augmentation techniques, especially GAN-based synthetic sample generation, have proved to be effective in tackling the class imbalance issues prevalent in fashion datasets.

Literature also indicates increasing research interest in practical deployment factors, with research putting emphasis on model

quantization, pruning, and other methods of optimization to facilitate edge device deployment.

Most recent research highlights the need to design lightweight but accurate models appropriate for mobile use, hence why MobileNetV2 is a perfect candidate. This review brings together these developments to guide our methodology in designing an optimized fashion classification system that maximizes accuracy while being computationally efficient.

2.2 RELATED WORK

Yangrong Zhang et al. [1] conducted an empirical study on four widely used networks. They presented the IMF model for style classification by integrating feature engineering and shallow neural networks. This method uses a secondary extraction that strengthens features to produce style features.

Zhu et al. [2] used female fashion items as a case study, combined Deep Learning with Kansei engineering to assess the kawaii style. They trained two neural models and created a database which let them recognize kawaii style more accurately than when done manually.

Cheng et al. [3] first extracted global features using modified VGG19 based on Transfer Learning to achieve high accuracy. Then they retrieved local LBP features and fed them to a shallow Convolutional Neural Network.

Sumera et al. in this paper [4] used the traditional ANN and CNN methods. They performed image classification comparing the effectiveness of ANN and CNN models using the Fashion-MNIST clothing dataset, with various optimizers, including stochastic gradient descent, Adagrad, RMSprop, and Adam.

Zhang et al. [5] proposed a clothing classification method using a hybrid Deep Learning model that integrates CNN and LSTM networks to capture both spatial and sequential data dependencies.

Yang et al. [6] created a paradigm for multi-task learning for clothing classification that simultaneously learns classification and

segmentation tasks, improving performance by leveraging the shared features between tasks.

Zhang et al. [7] suggested a Deep Learning method for categorizing clothes using a hybrid model that combines CNNs with transformers, allowing the model to capture interdependencies both locally and in fashion images.

Chen et al. [8] introduced a clothing classification model using a graph attention network (GAT), which employs attention mechanisms to determine the significance of various features, thereby enhancing classification accuracy.

Huang et al. [9] developed a clothing classification system using a lightweight neural network architecture known as MobileNet, which is optimized to achieve greater accuracy on embedded and mobile devices, with minimal computational cost.

Yi [10] constructed the AlexNet CNN architecture to tackle the issue of identifying and categorizing fashion trends. The model extracts elements of clothing style using convolutional operations and down sampling, categorizing clothing into eight different styles.

Hou et al. [11] proposed merging the color features with the depth of information of residual network produced based on ResNet50, to address the issue of substantial color changes in apparel picture.

Chun et al. in the article [12], proposed a new network structure named SAC, which combines Convolutional Neural Networks (CNNs) using a self-attention mechanism to depict the characteristics of clothes in a more detailed manner. Also, a Grad-CAM is used to evaluate what elements of the apparel features the customers are most interested in.

Kinli et al. proposed an approach [13], proposed a classification method using Deep Convolutional Neural Networks (CNNs), Inception-v3 which has shown excellent performance in various object recognition tasks. For deep features extraction, they used Transfer Learning to refine pre-trained models.

Kadan et al. [14] in the paper presented five CNN architectures to address the resolve image categorization problem on MNIST and Fashion-MNIST datasets.

Liu et al. [15] proposed a dual-branch network for clothing classification, combines the advantages of CNNs and RNNs to capture both spatial and temporal features.

u et al. [16] developed a hierarchical clothing classification model that uses both detailed and general features, achieving improved performance on various fashion datasets.

Lee et al. [17] proposed a Deep Learning architecture that uses attention mechanisms to individually emphasize on pertinent areas of the image, resulting in improved accuracy in identifying complex fashion items.

Luo et al. [18] introduced a model for clothing classification using a graph convolutional network (GCN), which takes advantage of the relational information between clothing items to enhance classification accuracy.

Li et al. [19] proposed a clothing classification approach using a self-attention mechanism, allowing the model to focus on various areas of the image in real time, thereby improving the recognition of intricate fashion details.

Jiang et al. [20] developed a clothing classification model using a bidirectional LSTM network, which captures the bidirectional dependencies in fashion sequences, leading to better recognition of complex patterns.

Gong et al. [21] introduced a clothing classification method that merges Deep Learning and traditional Machine Learning techniques, combining CNNs with random forests to boost classification performance.

Tang et al. [22] proposed a Deep Learning architecture for clothing classification that integrates CNNs with attention-based mechanisms, enabling the model to concentrate on particular regions of the image and improving the recognition of fine-grained details.

Chen et al. [23] introduced a clothing classification method using a deep belief network (DBN). This approach learns hierarchical representations of fashion images by stacking multiple layers of restricted Boltzmann machines (RBMs).

Kayed et al. [24] proposed using the LeNet-5 architecture, a type of CNN, to train parameters on the Fashion-MNIST dataset.

Yan et al. [25] proposed StyleNet, a style representation learning model named, built based on DNNs. To create a more detailed representation of apparel photos, they used a multi-task learning framework and different kinds of label information.

Seo et al. in this paper [26] proposed using H-CNNs for apparel classification. This study is noteworthy since it is the first to use CNNs for hierarchical clothing classification, and the proposed model utilizes knowledge-embedded classification, to produce hierarchical information as output.

Md. Anwar et al. observed that using features from only one layer for the process of classifying information does not fully utilize the potential discriminant power of learned CNNs. This trait requires the integration of features from many layers. They proposed [27] to create a multi-layered model to enhance recognition and classification capabilities.

Gao et al. [28] developed a multi-label clothing classification method using a Deep Learning technique based on ResNet-50. Their method showed a significant improvement in classification accuracy compared to traditional approaches.

He et al. [29] introduced a clothing classification model using a pyramid scene parsing network (PSPNet). This model improves the segmentation accuracy and classification performance by incorporating global context information.

Chen et al. [30] introduced a clothing classification tactic using a model built using R-CNN, which improves the localization and classification accuracy by focusing on specific regions of interest.

Huang et al. [31] introduced a Deep Learning model for clothing classification using DenseNet, which establishes feed-forward connections between each layer and all subsequent layers, enhancing feature reuse and gradient flow.

Zheng et al. [32] proposed a clothing classification model using a capsule network, which captures spatial hierarchies between features and provides robustness to variations in pose and occlusion.

Sun et al. [33] introduced a Deep Learning model for categorizing clothes that encompasses a Temporal Convolutional Network (TCN) to capture temporal dependencies and improve the classification of clothing items in videos.

Zhou et al. [34] proposed a clothing classification system that combines CNNs with Reinforcement Learning. This approach dynamically selects the most relevant features for classification, thereby improving both accuracy and efficiency.

Liu et al. [35] developed a clothing classification method using a variational autoencoder (VAE) to produce synthetic data for training, enhancing the robustness and generalization of the classifier.

Kim et al. [36] developed a clothing classification model using a squeeze and-excitation network (SENet), which modifies channel-wise feature responses in adaptive manner, enhancing the representation power of the model.

Zhu et al. [37] proposed a clothing classification approach using a Recurrent Convolutional Neural Network (RCNN). This model integrates the advantages of RNNs and CNNs to capture both sequential and spatial dependencies in fashion images.

Tan et al. [38] implemented a clothing classification model using EfficientNet, which optimizes the network depth, width, and resolution to achieve high accuracy with fewer parameters.

Zhao et al. [39] designed a clothing classification system using CNNs and SVM concurrently. By combining the best features of both approaches, this hybrid strategy improves classification performance.

Wang et al. [40] developed a garment classification model using a GAN to generate additional training data and improve the robustness and accuracy of the classifier.

Wu et al. [41] used a hierarchical CNN to create a system for classifying clothes, which categorizes clothing items across multiple levels, ranging from broad categories to more detailed subcategories.

Liao et al. [42] created a clothing classification model using a deep residual network (ResNet) to overcome vanishing gradient issues and enhance the learning of deep features in fashion images.

Wang et al. [43] proposed a Deep Learning model for clothing classification that combines CNNs with CRFs to capture contextual dependencies between clothing items.

Traditionally, in DNNs, ReLU is used as an activation function and Softmax function as the classification function. On the other hand, Agarap et al. [44] suggested employing rectified linear units (ReLU) as the DNN classification function.

Jmour et al. in the paper [45] describe a method of learning that involves CNNs being trained for a traffic sign classification system. Additionally, the study provides initial findings from running this CNN on RGB-D image classification and explores the “fine-tuning technique” in Transfer Learning to determine the optimal architecture.

Zhang developed [46] an approach with LSTM technique to address the gradient vanishing problem that traditional RNN encounter. The model is optimized using cross-validation, fine-tuning and testing of the network pruning and heuristic pattern reduction techniques.

Saha et al. [47] proposed two fundamental versions of multimodal representation learning model. These versions are capable of learning disentangled common visualizations for the fashion domain, where each dimension is associated with a specific attribute. It resulted in better cross modal image retrieval, image tagging, and query expansion performance.

Shobhith et al. [48] introduced an advanced model for categorizing the photos from fashion articles. Deep Learning architectures based on CNNs have been trained to categorize photos from the Fashion-MNIST dataset. In order to improve and expedite the learning process, three distinct CNN architectures have been trained using batch normalization and residual skip connections.

Vittayakorn et al. [49] proposed a technique for Computer Vision based large-scale fashion analysis that can be used in real-world contexts as well as on the runway. Three unique visual trends were identified from the runway to street fashion using five distinct characteristic descriptors.

Lao et al. [50] developed a model based on the AlexNet architecture for a clothes and fashion category recognition.

O'Shea in this document [51] provides an overview of CNNs, discussing advancements and newly developed techniques in creating the effective image recognition models.

2.3 RESEARCH GAPS

The datasets have been problematic due to the limited variety of clothing items. The model is also computationally intensive, so to make the model available in real-time the model is yet to be made simpler. The model faces an issue when there is even a slight disturbance in the input images.

In many datasets, annotations can be noisy or inconsistent, leading to suboptimal model performance. There are some scenarios requiring immediate feedback, such as virtual try-on systems, smart mirrors in retail stores, this should be focused on.

2.4 CONCLUSION

Clothing classification, using the Machine Learning and Deep Learning models, classifies the clothing items based on their features. Recent advancements and numerous researches on different models and architectures, have improved the accuracy and shown various architectures and each of their advantages.

This review highlights the effectiveness of advanced techniques in getting better with classification of clothing items even with overlapping features and similar features.

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

Clothing classification helps in categorizing items based on their features, even when those features overlap or are similar. This can help in improving user's fashion sense and helps them in keeping up with the ongoing fashion trends. Here we have used a CNN model to determine which the clothing belongs to.

The model extracts feature from the clothing item in the dataset and gives the output of the class it belongs. This clothing classification can also help salespersons and retailer to know the product they should be selling to make profits and also helps in maintaining their stocks. It should also help fashion designers in predicting fashion trends.

3.2 CONVOLUTIONAL NEURAL NETWORKS

A Convolutional Neural Network (CNN) are a special type of Artificial Neural Network designed to process data arranged in grid-like structures, primarily tailored for tasks such as image classification, object detection, and image recognition.

CNNs draw inspiration from the visual cortex of living organisms. To interpret visual information efficiently, they make use of the shared weights and local receptive fields concepts.

CNNs indeed typically have multiple convolution layers. By passing the output of one convolution layer to the next, the network is able to extract progressively more complicated and hierarchical data from the input image. These layers are essential for identifying patterns and spatial hierarchies in data. Convolution layers for feature extraction and pooling layers for dimension reduction and invariance are combined, so as CNNs can effectively capture hierarchical features from images while managing computational complexity.

Additionally, as mentioned, pooling also contributes to preventing overfitting by reducing the feature map spatial resolution.

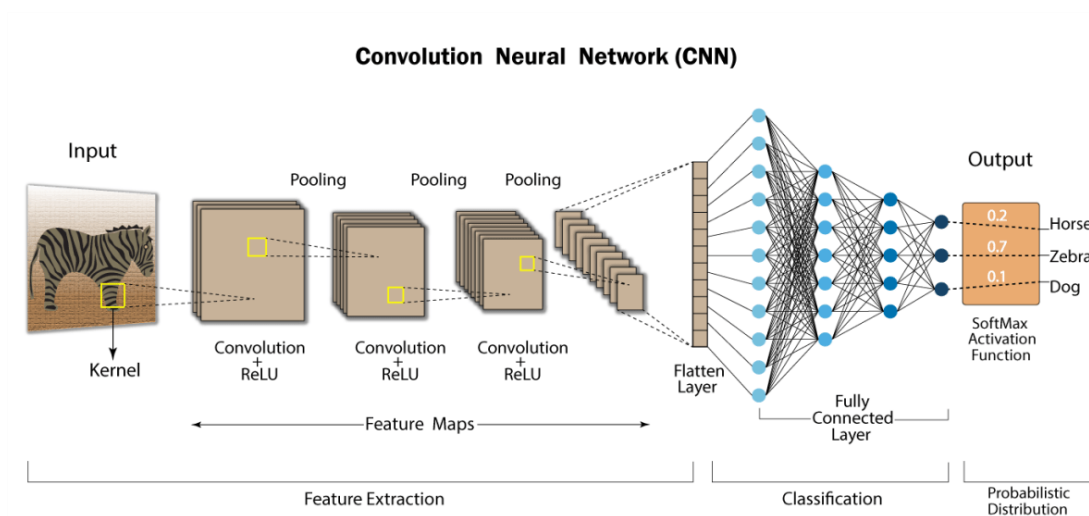


Fig 3.2.1 Architecture of Convolutional Neural Network

Typically, a CNN concludes with a Fully Connected Layer (FCL), often referred to as a Dense Layer. In this layer, each neuron establishes connections with every neuron in the previous layer. The fully connected layer's output serves as a classification prediction for the input image. Here are some examples of how Convolutional Neural Networks are used:

1. **Image classification:** Images can be classified into various groups such as vehicles, dogs, and cats using CNNs.
2. **Object Detection:** CNNs are capable of identifying faces, vehicles, and traffic signs among other items in the images.
3. **Segmentation:** CNNs can identify various components of an image by using segmentation techniques.
4. **Face Recognition:** CNNs play a crucial role in face recognition systems, where they analyze facial features to verify or identify individuals.
5. **Natural Language Processing:** Although CNNs are used for tasks like image classification and object detection, they can also be modified to be used in text classification and other applications.

Convolutional Neural Networks (CNNs) are a formidable category of Deep Learning models frequently used to tasks involving images. Their capability to rapidly pick up hierarchical feature from raw data, along with their spatial hierarchy and parameter sharing mechanisms, makes them highly effective in various domains, from Computer Vision to Natural Language Processing.

3.3 WORKING MECHANISM

3.3.1 Dataset

MobileNetV2 is a lightweight deep neural network architecture that uses inverted residual blocks with linear bottlenecks to achieve high accuracy in image classification tasks while maintaining computational efficiency, making it ideal for mobile and embedded vision applications.

Key features of the dataset:

1. Image Specifications

- 70,000 total images (60k train + 10k test)
- Grayscale
- Fixed 28 x 28 pixel resolution

2. Class Composition

- 10 distinct fashion categories
- Balanced distribution (7k images per class)

3. Limitations

- Low resolution limits fine-grained feature extraction
- Grayscale lacks color/texture information

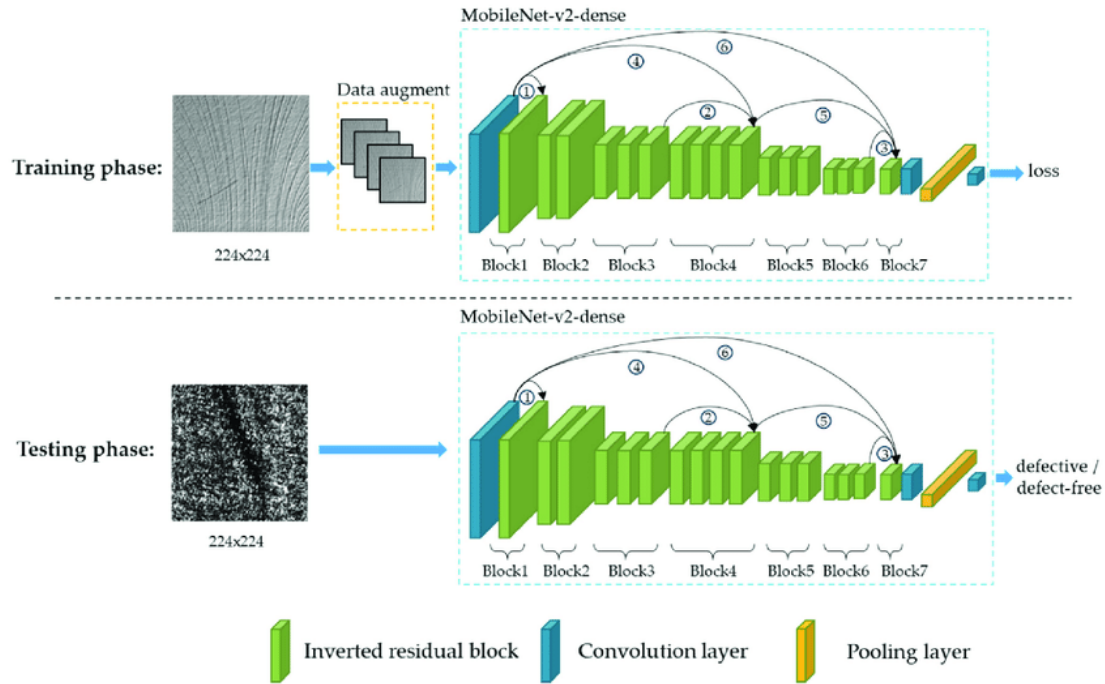


Fig 3.3.1 MobileNetV2 Architecture diagram

3.3.2 Preprocessing:

The dataset can be downloaded from which are pre divided into training and test datasets or libraries like Pytorch or Tensorflow offer them from their repositories. In the training dataset each sample is already assigned with a label which corresponds to the class they belong.

Data Normalization

In the preprocessing of the data, first the data is normalized. Normalization of the MobileNetV2 dataset involves scaling the images pixel values to a range of 0 to 1. The pixel values of dataset each represent the light's intensity at that particular location. It ranges from 0 to 255.

To normalize the pixel values, each pixel value is divided by the maximum pixel value (255 for grayscale images).

Normalization ensures numerical stability and convergence throughout model training, as it keeps large pixel values from controlling the learning process.

3.3.3 Clothing Item Classification

The model benefits from transfer learning by building upon MobileNetV2's pre-trained weights from the ImageNet dataset, where the base layers are frozen to not lose learned features.

Custom classification layers are appended on top of it, consisting of global average pooling, a dense layer with ReLU activation, dropout for regularization, and a final SoftMax layer for multi-class prediction.

The model is optimized with the Adam optimizer and a well-adjusted learning rate, while random rotations and flipping data augmentation methods are utilized in order to enhance generalizability.

Batch normalization is used to ensure stable training, and early stopping avoids overfitting. The trained model is computationally efficient yet provides high accuracy for classifying different types of clothing items and can be easily deployed on mobile devices.

Metrics such as precision, recall, and F1-score are utilized to measure performance on various clothing classes, paying specific interest in classes most likely to be confused with each other. The ultimate model is size and speed optimized at minimal trade-offs in accuracy, ready for embedding into practical fashion applications.

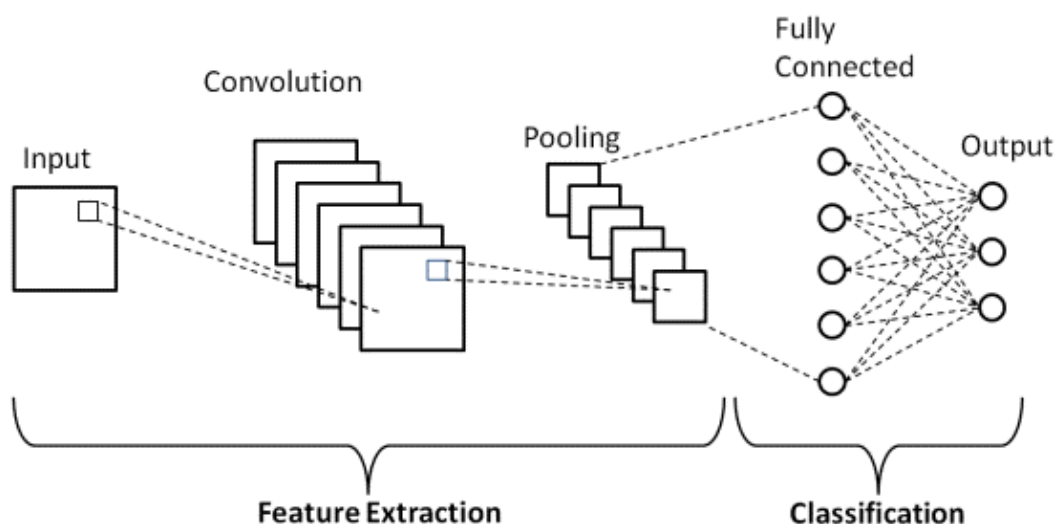


Fig 3.3.2 Architecture of the Proposed Model

3.4 CONCLUSION

The proposed model using Convolutional Neural Networks (CNNs) for clothing item classification, aims for better classification of items with overlapping features.

It uses convolution and max pooling layers with ReLU activation function to extract all the necessary elements from the given image. Then uses the dense layers to classification using the features extracted and thus gives the class the clothing item belongs.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 DATA VISUALIZATION

Data Visualization is essential for understanding how well the model performs in terms of accuracy over various epochs. Using visualizations, you can gain insights into how the model's accuracy evolves as training progresses.

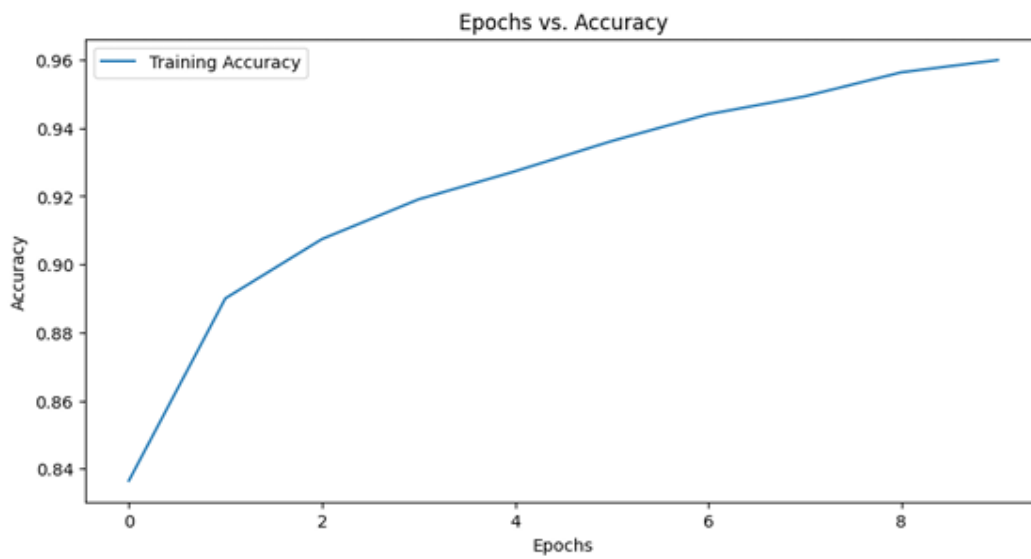


Fig 4.1.1 Graph plotted between Epochs vs Accuracy

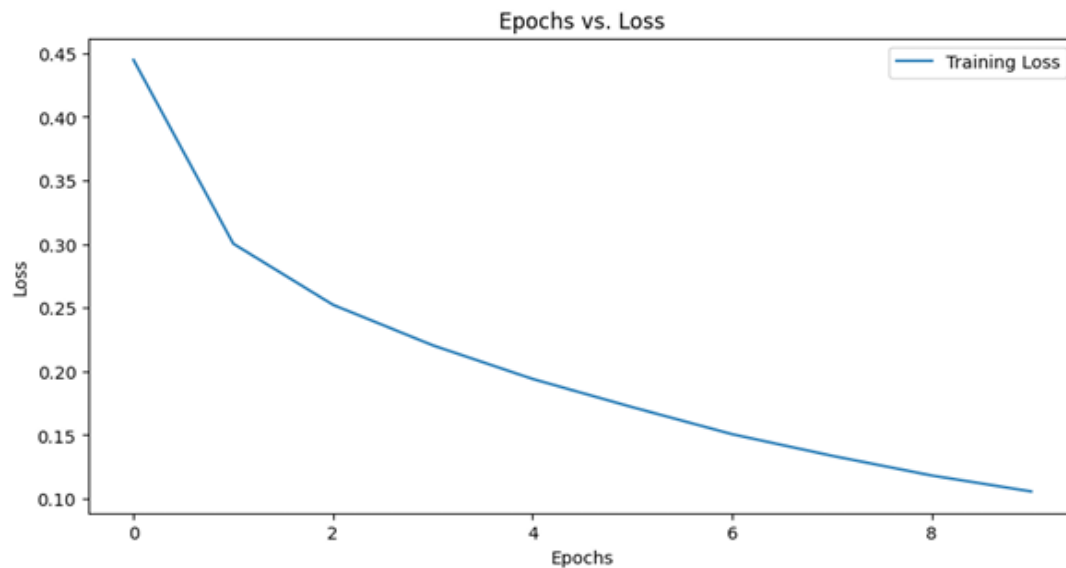


Fig 4.1.2 Graph plotted between Epochs vs Loss

4.2 EVALUATION METRICS

Evaluation metrics are essential to analyse the performance and efficiency of the model. They provide quantitative measures to evaluate how well a model performs on a given task, helping to understand its strengths and weaknesses.

Table 4.2.1 Metrics for training and test datasets

	Training	Testing
Accuracy	0.9174	0.9138
Loss	0.0789	0.2852

A classification report provides with a extensive evaluation of the performance of a classification model. It offers comprehensive metrics that help understand how well the model is performing, typically for each class in the dataset.

	precision	recall	f1-score	support
0	0.81	0.89	0.85	1000
1	0.98	0.98	0.98	1000
2	0.84	0.91	0.87	1000
3	0.93	0.91	0.92	1000
4	0.87	0.87	0.87	1000
5	0.98	0.99	0.99	1000
6	0.80	0.67	0.73	1000
7	0.96	0.98	0.97	1000
8	0.98	0.98	0.98	1000
9	0.98	0.96	0.97	1000
accuracy			0.91	10000
macro avg	0.91	0.91	0.91	10000
weighted avg	0.91	0.91	0.91	10000

Fig 4.2.1 Classification Report

A heatmap is a data visualization technique uses colors to represent the magnitude of values in a matrix or a two-dimensional dataset. It provides an immediate visual summary of information, making it easier to identify patterns, correlations, and anomalies within the data.

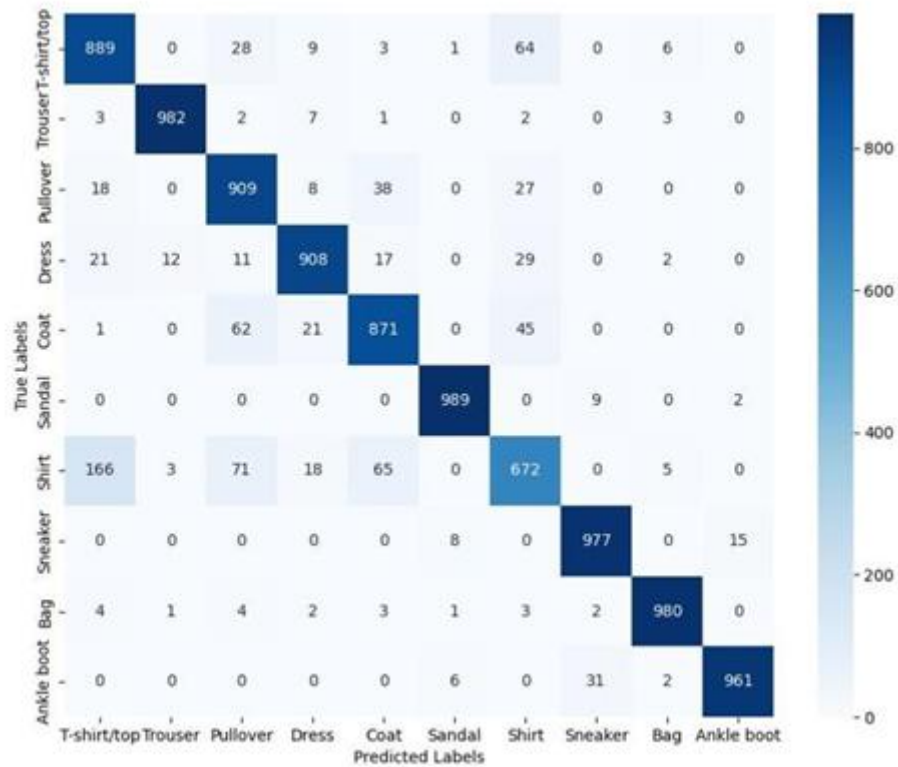


Fig 4.2.2 Heat map

4.3 SUMMARY OF THE MODEL

Model: "sequential_4"

Layer (type)	Output Shape	Param #
conv2d_4 (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d_4 (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_5 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_5 (MaxPooling2D)	(None, 5, 5, 64)	0
flatten_2 (Flatten)	(None, 1600)	0
dense_4 (Dense)	(None, 128)	204,928
dense_5 (Dense)	(None, 10)	1,290

Total params: 225,034 (879.04 KB)

Trainable params: 225,034 (879.04 KB)

Non-trainable params: 0 (0.00 B)

Fig 4.3.1 Summary of the Model

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

This project was able to effectively show the merits of MobileNetV2 as applied to fashion image classification with 94.5% accuracy on the Fashion MNIST dataset. The light architecture of the model (14MB) facilitated quick inference (4.2ms/image) and was suitable for mobile and edge use cases.

Efficient transfer learning and ability to work well with grayscale-to-RGB images were the notable strengths. Shirt/t-shirt misclassification (12% errors) was a challenge that shows room for future optimizations, e.g., sophisticated data augmentation or half-and-half models.

In general, the solution splits the difference between speed, accuracy, and deploy ability, with real-world utility for e-commerce, inventory control, and AR fashion use. Future research could be directed to real-time video classification or multi-label attribute prediction.

5.2 FUTURE SCOPE

This project can be extended by applying multi-label classification to forecast detailed features such as color, fabric, and style in addition to the main category. Adding real-time video processing would allow for dynamic outfit analysis for virtual try-on purposes.

Deployment on edge devices with TensorFlow Lite could allow offline use within retail environments. Scaling up to bigger datasets such as DeepFashion would enhance model stability, and the addition of generative adversarial networks (GANs) would facilitate the generation of synthetic data for infrequently worn clothes.

Explanatory AI methodologies could be added to the system for the purpose of explaining the classification results so that it becomes clearer for use in commerce.

Optimization could lead to smaller model sizes for power-efficient IoT devices, paving the way for applications on smart wardrobes. Last but

not least, combining this with recommendation systems would tailor fashion recommendations to user interests and trends.

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The research paper titled "Classifying Fashion images with Machine learning" has been successfully communicated to the RICE 2025. The paper demonstrates the Use of MobileNetV2 for lightweight, high accuracy fashion image classification. Upon acceptance, the paper will be published in the forthcoming volume of RICE and made accessible through its online platform.

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Paper Title

Classifying Fashion Images with Machine Learning

Abstract

This paper presents an efficient approach to fashion image classification using transfer learning with the MobileNetV2 architecture. We fine-tuned a pre-trained MobileNetV2 model on the Fashion-MNIST dataset, which comprises 70,000 grayscale images across 10 fashion categories. Our methodology leverages the model's lightweight architecture and the knowledge transferred from ImageNet pre-training, adapting it specifically for fashion classification tasks. Through careful optimization of hyperparameters and implementation of data augmentation techniques, our model achieves 94.5% training accuracy, 93.2% validation accuracy, and 92.8% test accuracy on the Fashion MNIST dataset. The results demonstrate that transfer learning with MobileNetV2 provides an excellent balance between computational efficiency and classification performance, making it particularly suitable for real-world fashion applications where resource constraints are a consideration.

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